

Software Test Plan (STP)

Student Attendance Management System

Item	Details
Project	Student Attendance Management System
Version	1.0
Authors	Bindu S , Sohan MR Srinikesh Durgavajjula Taanya Gupta
Date	28-09-2026
Status	For Review

1. Introduction

Purpose: This document defines the test plan for the Student Attendance Management System (SAMS) v1.0. It specifies the testing objectives, scope, strategy, resources, schedule, responsibilities, risks, and reporting approach.

Scope: Testing covers authentication, student management, faculty management, subject/class management, attendance marking and updating, attendance history, percentage calculation, reports, alerts, and role-based access. Institutional infrastructure and external systems outside the application boundary are excluded.

References: Student Attendance Management System SRS v1.0, system design specifications, database design, institutional attendance rules, and project acceptance criteria.

Definitions: SAMS (Student Attendance Management System), Admin, Faculty, Student, Attendance Session, Attendance Percentage, FR (Functional Requirement), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), UAT (User Acceptance Testing).

2. Test Items

- User authentication and role-based access module
- Student registration and record management module
- Faculty management module
- Subject and class management module
- Attendance session creation and marking module
- Attendance update/correction module
- Attendance history and percentage calculation module
- Attendance reports and low-attendance identification module
- Administration and configuration module
- Audit/logging and notification functions

3. Features to be Tested

- SAMS-F-001: Secure login and role-based authentication
- SAMS-F-002: Student registration and record management
- SAMS-F-003: Faculty account and profile management
- SAMS-F-004: Subject/class creation and assignment
- SAMS-F-005: Create attendance session

- SAMS-F-006: Mark student attendance
- SAMS-F-007: Update/correct attendance with authorization
- SAMS-F-008: View attendance history
- SAMS-F-009: Calculate attendance percentage
- SAMS-F-010: Generate attendance reports
- SAMS-F-011: Identify students below configured attendance threshold
- SAMS-F-012: Student attendance dashboard/view
- SAMS-F-013: Faculty attendance dashboard
- SAMS-F-014: Admin user/role management
- SAMS-F-015: Search/filter attendance records
- SAMS-F-016: Audit important attendance changes
- SAMS-F-017: Attendance rules/configuration
- SAMS-F-018: Logout/session termination
- SAMS-NF-001 to SAMS-NF-007: Performance, availability, usability, reliability, maintainability, scalability, and compatibility requirements
- SAMS-SR-001 to SAMS-SR-006: Security requirements

4. Features Not to be Tested

- College ERP or examination-system functionality outside the attendance module.
- Physical biometric devices or RFID hardware unless explicitly included in the project scope.
- Operating-system internals and underlying database engine internals.
- Third-party cloud/network infrastructure outside the application boundary.
- Features explicitly excluded from the v1.0 SRS.

5. Test Approach / Strategy

Levels: Unit testing, integration testing, system testing, and user acceptance testing (UAT).

Types: Functional, integration, regression, performance/load, usability, compatibility, negative/boundary, security, and data-validation testing.

Entry Criteria: Stable build available; database and test accounts prepared; test environment configured; attendance rules and sample academic data available; reviewed test cases ready.

Exit Criteria: 100% of planned test cases executed; no open critical defects; mandatory acceptance criteria satisfied; major defects resolved or formally accepted; final test summary completed.

5.1 Security Validation

- Validate login authentication and password handling.
- Verify role-based access for Admin, Faculty, and Student users.
- Verify students cannot modify their own attendance records.
- Verify only authorized faculty/admin users can mark or correct attendance.
- Test invalid input, unauthorized requests, and session-expiry behavior.
- Verify sensitive user information and passwords are not stored or logged in plain text.
- Verify audit records are generated for important attendance changes.

6. Test Environment

Hardware: Developer/test laptops or desktops connected through the college/local network or controlled Internet environment.

Software: Student Attendance Management System v1.0, web browser, application server, relational database, and required runtime.

Tools: Browser developer tools, Postman or equivalent API tool where applicable, database client, GitHub for version control, and Jira/GitHub Issues for defect tracking.

Test Data: Dummy student/faculty accounts, subjects, classes, attendance sessions, present/absent records, boundary percentages, invalid credentials, and sample reports.

7. Test Schedule

Milestone	Planned Date
Test case design	01-Oct-2026
Test environment setup	03-Oct-2026
Test execution start	05-Oct-2026
Functional/integration testing end	12-Oct-2026
Performance & security testing	14-Oct-2026
UAT	15-Oct-2026 to 17-Oct-2026
Final test report	19-Oct-2026

8. Test Deliverables

- Software Test Plan
- Test Cases
- Test Scripts / Automation Scripts where applicable
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Taanya Gupta	Prepare test plan, coordinate testing, review results.
Test Engineer	Bindu S	Design and execute test cases, record evidence, and log defects.
Developer	Sohan MR	Support defect analysis, fixes, builds, and technical clarification.
Product Owner	Srinikesh Durgavajjula	Clarify requirements, review acceptance results, and approve readiness.

10. Risks and Mitigation

Risk	Mitigation
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Unstable application build	Request early smoke builds and maintain a known stable build.
Incorrect attendance calculations	Use independent sample calculations and boundary-value test data.
Unauthorized attendance modification	Perform RBAC and authorization testing for every attendance operation.
Database/data loss	Maintain backup test data and verify database backup/recovery procedures.
Late defect fixes	Prioritize critical defects and perform regression testing after fixes.

11. Assumptions & Dependencies

- Student, faculty, subject, and class test data will be available before execution.
- The attendance rules and minimum percentage threshold will be provided by the project/institution.
- The application server and database will be available during scheduled testing.
- Developers will provide stable builds and defect fixes for retesting.
- Test artifacts and source-code contributions will be maintained in GitHub.

12. Suspension & Resumption Criteria

Suspend testing if: The application/test environment is unavailable for more than 4 hours; a build blocks more than 30% of planned test cases; or a critical database/application failure prevents meaningful execution.

Resume testing if: Blocking defects are resolved or an approved workaround is available, the environment is stable, and a new build passes smoke testing.

13. Test Case Management & Traceability

RTM: The Requirements Traceability Matrix maps SRS requirements to test cases so that every important requirement is verified.

Requirement	Example Test Cases
SAMS-F-001 – Login/RBAC	TC-AUTH-01, TC-AUTH-02, TC-AUTH-03
SAMS-F-006 – Mark attendance	TC-ATT-01, TC-ATT-02, TC-ATT-03
SAMS-F-007 – Update attendance	TC-ATT-04, TC-ATT-05
SAMS-F-009 – Attendance percentage	TC-CALC-01, TC-CALC-02
SAMS-F-010 – Reports	TC-REP-01, TC-REP-02
SAMS-F-011 – Low attendance	TC-ALERT-01, TC-ALERT-02
SAMS-NF-001 – Performance	TC-PERF-01, TC-PERF-02
SAMS-SR-001 – Authentication security	TC-SEC-01, TC-SEC-02

14. Test Metrics & Reporting

- Percentage of test cases executed
- Percentage passed, failed, and blocked
- Defect count by severity
- Defect density and defect aging
- Requirement coverage
- Regression pass rate
- Average response time for major operations

- Attendance calculation accuracy

Reports: Daily test execution status, defect status report, and final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA Lead	Tanya Gupta	
Dev Lead	Sohan MR	
Product Owner	Srinikesh Durgavajjula	